

Chapter 26: Resolvents & Fixed Points

Zeros of Sums of Monotone Operators

Convex Analysis and Monotone Operator Theory

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Chapter Overview

- **Problem:** Find zeros of sums of monotone operators

Find $x \in \mathcal{H}$ such that $0 \in Ax + Bx$

- **Key Concept:** Resolvents and Fixed Points

- Resolvent: $J_A = (\text{Id} + A)^{-1}$
- Fixed point: $x = J_A(x) \Leftrightarrow 0 \in Ax$

- **Main Topics:**

- 1 Duality and Characterizations
- 2 Spingarn's Method of Partial Inverses
- 3 Douglas-Rachford Splitting Algorithm
- 4 Forward-Backward Splitting
- 5 Variational Inequalities

- **Applications:** Optimization, proximal methods, splitting algorithms

Zeros of Sums: Basic Characterizations

Setup: $A, B : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ are operators, $\gamma \in \mathbb{R}_{++}$

Let $\mathcal{P} = \text{zer}(A + B)$ (solutions to $0 \in Ax + Bx$)

Let $\mathcal{Q} = \text{zer}(-A^{-1}(-\text{Id}) - B^{-1})$ (dual problem)

Proposition 26.1: The following characterizations hold:

(i) **Direct form:**

$$\mathcal{P} = \{x \in \mathcal{H} \mid (\exists u \in \mathcal{Q}) -u \in Ax \text{ and } u \in Bx\}$$

$$\mathcal{Q} = \{u \in \mathcal{H} \mid (\exists x \in \mathcal{P}) x \in A^{-1}(-u) \text{ and } x \in B^{-1}u\}$$

(ii) **With closed affine subspace:** If C is closed affine and $A = N_C$:

$$\mathcal{P} = \{x \in C \mid V^{\perp} \cap Bx \neq \emptyset\}, \quad \mathcal{Q} = \{u \in V^{\perp} \mid C \cap B^{-1}u \neq \emptyset\}$$

(iii) **Douglas-Rachford operator:** Let $T_{A,B} = J_A R_B + \text{Id} - J_B$

$$\mathcal{P} = J_{\gamma B}(\text{Fix } R_{\gamma A} R_{\gamma B}), \quad \mathcal{Q} = {}^{\infty} B(\text{Fix } T_{\gamma A, \gamma B})$$

Douglas-Rachford Splitting Operator

Definition: For maximally monotone A, B , the **Douglas-Rachford splitting operator** is:

$$T_{A,B} = J_A R_B + \text{Id} - J_B$$

where:

- $J_A = (\text{Id} + A)^{-1}$ is the resolvent
- $R_A = 2J_A - \text{Id}$ is the reflected resolvent

Key Properties (Proposition 26.1(iii)):

- (a) $T_{A,\gamma B}$ is firmly nonexpansive
- (b) $\mathcal{P} = J_{\gamma B}(\text{Fix } R_{\gamma A} R_{\gamma B}) = J_{\gamma B}(\text{Fix } T_{\gamma A, \gamma B})$
- (c) $\mathcal{Q} = {}^\infty B(\text{Fix } R_{\gamma A} R_{\gamma B}) = {}^\infty B(\text{Fix } T_{\gamma A, \gamma B})$
- (d) $\mathcal{P} \neq \emptyset \Leftrightarrow \mathcal{Q} \neq \emptyset \Leftrightarrow \text{Fix } T_{\gamma A, \gamma B} \neq \emptyset$

Special Cases

Proposition 26.1(iv): Suppose A is maximally monotone and $B : \mathcal{H} \rightarrow \mathcal{H}$. Set $T_{A,B} = J_A \circ (\text{Id} - B)$. Then:

(a) $\mathcal{P} = \text{Fix } T_{\gamma A, \gamma B}$

(b) $\mathcal{Q} = B(\mathcal{P})$

(c) $\mathcal{P} \neq \emptyset \Leftrightarrow \mathcal{Q} \neq \emptyset \Leftrightarrow \text{Fix } T_{\gamma A, \gamma B} \neq \emptyset$

Proposition 26.1(v): Suppose A is maximally monotone and $B^{-1} : \mathcal{H} \rightarrow \mathcal{H}$. Then:

$$\mathcal{Q} = \text{Fix}(-\gamma A \circ (B^{-1} - \gamma \text{Id})), \quad \mathcal{P} = B^{-1}(\mathcal{Q})$$

Corollary 26.3: For maximally monotone A and $\gamma \in \mathbb{R}_{++}$:

$$\text{Fix } J_A \circ (\text{Id} + \gamma(B - \text{Id})) = \text{Fix } J_{\gamma^{-1}A} \circ B$$

Finite Sums of Operators

Proposition 26.4: For a finite collection of maximally monotone operators $(A_i)_{i \in I}$, define:

$$\mathcal{H} = \bigoplus_{i \in I} \mathcal{H}, \quad D = \{(x, \dots, x) \in \mathcal{H} \mid x \in \mathcal{H}\}$$

$$\mathcal{J} : \mathcal{H} \rightarrow \mathcal{D}, \quad x \mapsto (x, \dots, x), \quad \mathcal{A} : \mathcal{H} \rightarrow 2^{\mathcal{H}}, \quad (x_i)_{i \in I} \mapsto \prod_{i \in I} A_i x_i$$

Then for $x = (x_i)_{i \in I} \in \mathcal{H}$:

- (i) $D^\perp = \{u \in \mathcal{H} \mid \sum_{i \in I} u_i = 0\}$
- (ii) $N_{Dx} = \begin{cases} \{u \in \mathcal{H} \mid \sum_{i \in I} u_i = 0\} & \text{if } x \in D \\ \emptyset & \text{otherwise} \end{cases}$
- (iii) $P_{Dx} = \mathcal{J}((1/m) \sum_{i \in I} x_i)$
- (iv) $P_{D^\perp x} = (x_i - (1/m) \sum_{j \in I} x_j)_{i \in I}$

Zeros of Finite Sums I

Theorem 26.5: Suppose $(A_i)_{i \in I}$ are maximally monotone operators where $I = \{1, \dots, m\}$ with $m \geq 2$. Then there exists $x \in \text{zer } \sum_{i \in I} A_i$ such that:

- (i) $x = x_1 = \dots = x_m$
- (ii) $\sum_{i \in I} u_i = 0$
- (iii) $(\forall i \in I) \quad (x, u_i) \in \text{gra } A_i$
- (iv) $\sum_{i \in I} \langle x_i, n | u_{i,n} \rangle \rightarrow \langle x | \sum_{i \in I} u_i \rangle = 0$

Proof sketch:

- Define $\mathcal{H}, \mathcal{D}, \mathcal{J}, \mathcal{A}$ as in Proposition 26.4
- Note that $(\mathbf{x}_n, \mathbf{u}_n)_{n \in \mathbb{N}}$ lies in $\text{gra } \mathcal{A}$
- Convergence follows from Proposition 23.18 and proximal-point iterations

Problem Setup

Spingarn's Method: A technique for solving constrained monotone operator problems.

Problem: Given:

- $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ is maximally monotone
- V is a closed linear subspace of $\mathcal{H}$
- The problem find $x \in V, u \in V^{\perp}$ such that $u \in Ax$ has at least one solution

Goal: Find $(x^*, u^*) \in V \times V^{\perp}$ such that $u^* \in Ax^*$

Application: Decompose a complex operator problem into subproblems that may be easier to solve.

Key decomposition: $\mathcal{H} = V \oplus V^{\perp}$

Spingarn's Algorithm

Theorem 26.9 (Spingarn): Let $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ be maximally monotone, V a closed subspace, and suppose the problem find $x \in V, u \in V^{\perp}$ such that $u \in Ax$ has at least one solution.

Given $x_0 \in V, u_0 \in V^{\perp}$, set for $n = 0, 1, \dots$:

$$\begin{aligned}y_n &= J_A(x_n + u_n) \\v_n &= x_n + u_n - y_n \\(x_{n+1}, u_{n+1}) &= (P_V y_n, P_{V^{\perp}} v_n)\end{aligned}$$

Then: There exists a solution (x, u) to the constrained problem such that $x_n \rightarrow x$ and $u_n \rightarrow u$.

Proof idea:

- Decompose $z_n = x_n + u_n$ where $z_n \in \mathcal{H}$
- Use proximal-point iterations: $z_{n+1} = J_A z_n$
- Project back to $V \times V^{\perp}$ at each step
- Convergence follows from Proposition 23.30(ii)

Spingarn's Method of Partial Inverses

Given: $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ maximally monotone, closed subspace V

Find: $x \in V, u \in V^{\perp}$ such that $u \in Ax$

Step 1: Initialize $x_0 \in V, u_0 \in V^{\perp}$

Step 2: Compute $y_n = J_A(x_n + u_n)$

Step 3: Compute $v_n = x_n + u_n - y_n$

Step 4: Project $(x_{n+1}, u_{n+1}) = (P_V y_n, P_{V^{\perp}} v_n)$

Standard Splitting Problem

Problem: Solve the monotone inclusion:

$$0 \in Ax + Bx$$

where $A, B : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ are maximally monotone operators.

Motivation:

- Direct methods may be hard to apply when A or B has no closed form
- **Splitting:** Apply resolvents of A and B separately
- Benefits: Each J_A and J_B may be easier to compute individually

Douglas-Rachford Splitting Algorithm:

$$\begin{aligned}y_n &= J_A(x_n) \\x_{n+1} &= x_n + J_B(2y_n - x_n) - y_n\end{aligned}$$

Equivalently: $x_{n+1} = T_{A,B}(x_n)$ where $T_{A,B} = J_A R_B + \text{Id} - J_B$

Fixed point: $x^* = T_{A,B}(x^*) \Rightarrow x^* \in \text{zer}(A + B)$

Convergence of Douglas-Rachford

Theorem 26.10: Suppose:

- (i) D is a nonempty closed subset of $\mathcal{H}$
- (ii) $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ is maximally monotone with $\text{dom } A \subseteq D$
- (iii) $B : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ is maximally monotone
- (iv) $\text{zer}(A + B) \cap D \neq \emptyset$

Given $x_0 \in D$, set:

$$\begin{aligned}y_n &= J_A(x_n) \\x_{n+1} &= x_n + J_B(2y_n - x_n) - y_n\end{aligned}$$

Then: $(x_n)_{n \in \mathbb{N}} \rightarrow x^*$ for some $x^* \in \text{zer}(A + B) \cap D$.

Proof sketch:

- Show x_n lies in bounded sequence (bounded domain)
- Use weak sequential compactness
- Apply monotonicity arguments to identify limit point

Douglas-Rachford Example

Numerical example: Solve $0 \in \partial f(x) + Bx$ where:

- $f(x) = \|x\|_1$ (sparsity-inducing)
- B is a linear operator

```
1 import numpy as np
2 from scipy.optimize import minimize
3
4 # Problem: min ||x||_1 + 0.5 * ||x - b||^2
5 def soft_threshold(x, lam):
6     """Proximal operator of ||.||_1"""
7     return np.sign(x) * np.maximum(np.abs(x) - lam, 0)
8
9 def douglas_rachford(b, gamma=0.1, n_iter=100):
10     """Douglas-Rachford for ||x||_1 + 0.5||x-b||^2"""
11     x = np.zeros_like(b)
12     for n in range(n_iter):
13         # Prox of ||.||_1
14         y = soft_threshold(x, gamma)
15         # Prox of 0.5||x-b||^2: (I + gamma*(-I)^{-1})^{-1}
16         z = (x + gamma * b) / (1 + gamma)
17         x = x + z - y
18     return x
```

Convergence Rates

Proposition 26.16 (Linear Convergence): Let:

- (i) A be α -strongly monotone, B be δ -cocoercive, $\gamma \in]0, 2\delta[$
- (ii) OR $\alpha \leq \beta$, B be α -strongly monotone and β -Lipschitz continuous, $\gamma \in]0, 2\alpha/\beta^2[$

Let $x_0 \in D$ and set:

$$\begin{aligned}y_n &= x_n - \gamma Bx_n \\x_{n+1} &= J_{\gamma A}(y_n)\end{aligned}$$

Then: $(x_n)_{n \in \mathbb{N}}$ converges linearly to the unique point in $\text{zer}(A + B)$.

Key rates:

- Linear: $\|x_{n+1} - x^*\| \leq \rho \|x_n - x^*\|$ with $\rho < 1$
- Rate depends on strong convexity/cocoercivity constants

Forward-Backward Algorithm I

Problem: Solve $0 \in Ax + Bx$ where:

- $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ is maximally monotone
- $B : \mathcal{H} \rightarrow \mathcal{H}$ is single-valued (e.g., differentiable operator)

Advantage: Can handle composite inclusion problems directly.

Forward-Backward Algorithm:

$$\begin{aligned}y_n &= x_n - \gamma Bx_n \quad (\text{forward step on } B) \\x_{n+1} &= J_{\gamma A}(y_n) \quad (\text{backward step on } A)\end{aligned}$$

Proposition 26.20: Suppose B is monotone and $\text{zer}(A + B) \neq \emptyset$, $\gamma \in]0, \infty[$. Then:

- (i) $(x_n)_{n \in \mathbb{N}}$ converges weakly to a solution
- (ii) If additionally A or B is strictly monotone, convergence is strong

Forward-Backward Algorithm II

Application to proximal gradient: Minimize $f(x) + g(x)$ where:

- f is differentiable (set $A = \partial f$)
- g is convex (set $B = \text{Id}$ in forward-backward, use ∂g for backward)

Resolvent Operator Properties

Definition: For $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ monotone, the **resolvent** is:

$$J_A = (\text{Id} + A)^{-1}$$

Fundamental Properties:

- ① **Existence:** If A is maximally monotone, J_A is defined on all of $\mathcal{H}$ and single-valued
- ② **Firmly Nonexpansive:** $\|J_A(x) - J_A(y)\|^2 + \|x - y - (J_A(x) - J_A(y))\|^2 \leq \|x - y\|^2$
- ③ **Nonexpansive:** $\|J_A(x) - J_A(y)\| \leq \|x - y\|$
- ④ **Fixed Points:** $\text{Fix}(J_A) = \text{zer}(A)$
- ⑤ **Lax-Milgram:** If $A = \partial f$ for $f \in \Gamma_0(\mathcal{H})$, then $J_A = \text{prox}_f$

Reflected Resolvent

Definition: The **reflected resolvent** is:

$$R_A = 2J_A - \text{Id}$$

Properties:

- ① $(R_A(x) - x)/2 = J_A(x) - x$ (average of input and output)
- ② R_A is also nonexpansive
- ③ For $x^* = J_A(x^*)$ (fixed point): $R_A(x^*) = x^*$
- ④ Composition $R_A \circ R_B$ has fixed points that solve the splitting problem

Connection to Douglas-Rachford:

$$T_{A,B} = J_A \circ R_B + \text{Id} - J_B$$

This operator mixes resolved and reflected operators to achieve splitting.

Fixed Point Iteration Theory

Proximal Point Algorithm:

$$x_{n+1} = J_{\lambda_n A}(x_n)$$

where (λ_n) is a sequence of positive stepsizes.

Theorem 26.11: If A is maximally monotone and $\text{zer}(A) \neq \emptyset$, and $\sum_n \lambda_n(\delta - \lambda_n) = +\infty$ for some $\delta > 0$, then:

- ① (x_n) converges weakly to some $x^* \in \text{zer}(A)$
- ② (Bx_n) converges strongly to Bx^* if B is single-valued

Corollary: With constant stepsize $\lambda_n = \lambda \in (0, \delta]$:

$$x_n \rightharpoonup x^* \in \text{zer}(A)$$

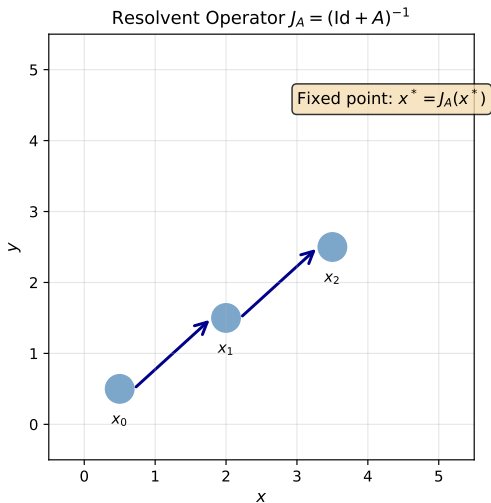
Interpretation: The proximal point algorithm monotonically approaches zeros of monotone operators without direct knowledge of the operator's structure.

Proximal Point Algorithm Example

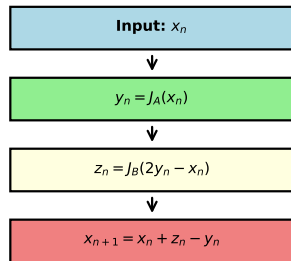
Example: Minimize $f(x) = \|x\|_1$ (proximal gradient descent)

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 def prox_l1(x, lam):
5     """Proximal operator of lambda ||.||_1"""
6     return np.sign(x) * np.maximum(np.abs(x) - lam, 0)
7
8 def proximal_point_algorithm(x0, lam=0.1, n_iter=50):
9     """Proximal point:  $x_{n+1} = \text{prox}_{\{\lambda f\}}(x_n)$ """
10    x = x0.copy()
11    errors = [np.linalg.norm(x)]
12
13    for n in range(n_iter):
14        x = prox_l1(x, lam)
15        errors.append(np.linalg.norm(x))
16
17    return x, np.array(errors)
18
19 # Test: minimize ||x||_1 from x0 = [2, -1, 0.5]
20 x0 = np.array([2.0, -1.0, 0.5])
21 x_final, errors = proximal_point_algorithm(x0, lam=0.05)
```

Visual Summary: Resolvents



Douglas-Rachford Splitting Operator



Repeat until convergence

Variational Inequality Problem

Definition: Given $C \subseteq \mathcal{H}$ convex and $B : \mathcal{H} \rightarrow 2^{\mathcal{H}}$, the **variational inequality** is:

$$\text{find } x \in C \text{ such that } (\forall y \in C) \quad \langle x - y \mid Bx \rangle \leq 0$$

Equivalence to monotone inclusions:

Let $A = N_C$ (normal cone). Then:

$$x \text{ solves VI} \Leftrightarrow 0 \in Ax + Bx \Leftrightarrow 0 \in N_C(x) + Bx$$

Splitting approach: Apply Douglas-Rachford or forward-backward to the equivalent inclusion.

Projection method:

$$x_{n+1} = P_C(x_n - \gamma Bx_n)$$

where P_C is the projection operator onto C .

Forward-Backward for Variational Inequalities

Proposition 26.25 (Forward-Backward Algorithm): Let $f \in \Gamma_0(\mathcal{H})$, $\beta \in \mathbb{R}_{++}$, $B : \mathcal{H} \rightarrow \mathcal{H}$ be β -cocoercive, $\gamma \in]0, 2\beta[$. Suppose:

$$\text{find } x \in \mathcal{H} \text{ such that } (\forall y \in \mathcal{H}) \quad \langle x - y \mid Bx \rangle + f(x) < f(y)$$

has at least one solution. Given $x_0 \in \mathcal{H}$ and $(\lambda_n) \subseteq [0, \delta]$ with $\sum_n \lambda_n(\delta - \lambda_n) = +\infty$:

$$\begin{aligned} y_n &= x_n - \gamma Bx_n \\ x_{n+1} &= x_n + \lambda_n(\text{prox}_f(y_n) - x_n) \end{aligned}$$

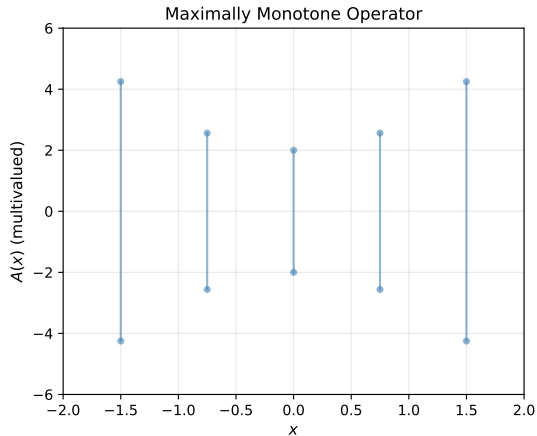
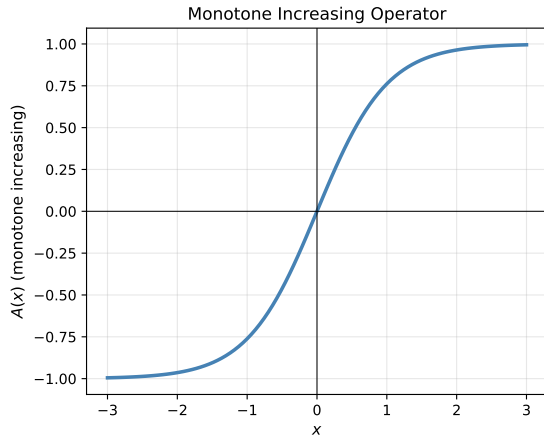
Then: (x_n) converges weakly to a solution, and if f or B is strictly convex, convergence is strong.

Variational Inequality Example

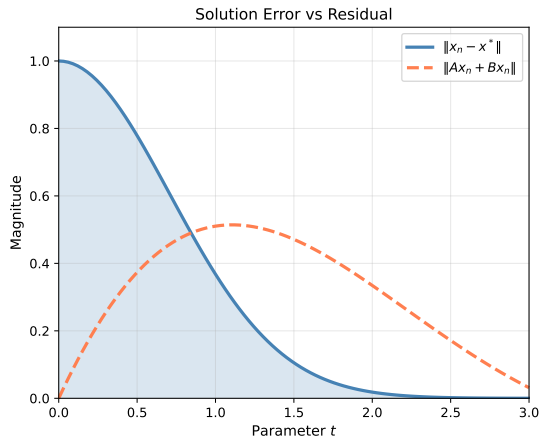
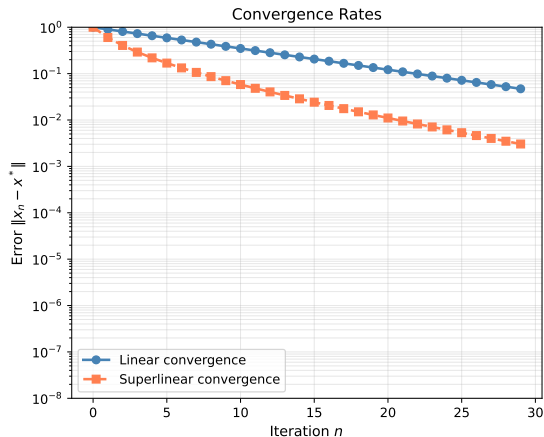
Example: Constraint qualification with quadratic penalty

```
1 import numpy as np
2
3 # Problem: min 0.5 ||Ax - b||^2 s.t. x in C = [0, 1]^n
4 # Equivalent VI: find x in C such that
5 #               <x-y, A^T(Ax-b)> <= 0 for all y in C
6
7 def projection_simplex(x, s=1.0):
8     """Project onto simplex: C = {x : sum(x)=s, x>=0}"""
9     n = len(x)
10    rho = np.sum(np.sort(x)[::-1]) - s
11    idx = np.arange(n) + 1
12    mask = np.sort(x)[::-1] - rho / idx > 0
13    theta = (np.sum(np.sort(x)[::-1][mask]) - s) / np.sum(mask)
14    return np.maximum(x - theta, 0)
15
16 def forward_backward_vi(A, b, x0, gamma=0.1, n_iter=100):
17     """Solve min ||Ax-b||^2 s.t. x in simplex"""
18     x = x0.copy()
19     for n in range(n_iter):
20         # Gradient of ||Ax-b||^2
21         grad_f = A.T @ (A @ x - b)
```


Monotone Inclusion Applications



Convergence Rates

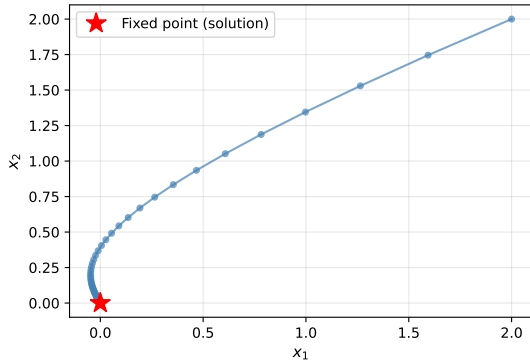


Key observations:

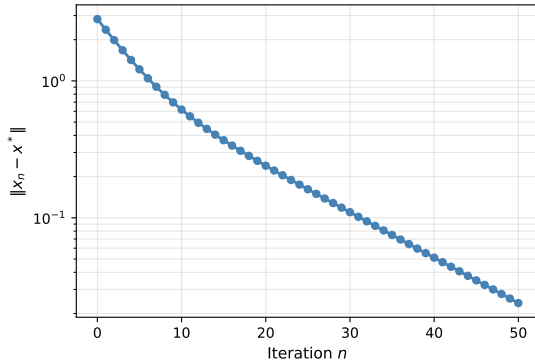
- **Linear convergence:** Achieved under strong monotonicity + cocoercivity
- **Superlinear convergence:** Possible with acceleration or additional structure
- **Residual vs distance:** Monotonic decrease in both error and residual

Numerical Example: Proximal Point Algorithm

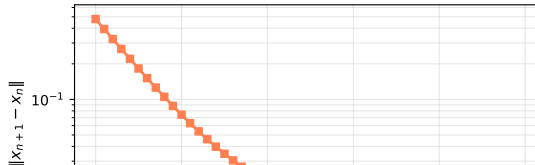
Algorithm Trajectory



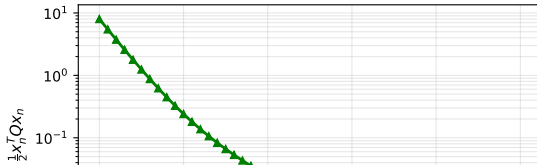
Error vs Iteration



Step Size Decrease



Objective Value



Fixed Point Theory Summary

Fixed Point Theory in Monotone Operator Theory

$A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ maximally monotone operator

$\Rightarrow$ **Resolvent** $J_A = (\text{Id} + A)^{-1}$ is **firmly nonexpansive**

Every fixed point of J_A is a zero of A

Nonexpansivity: $\|J_A(x) - J_A(y)\| \leq \|x - y\|$

Firmly nonexpansive: $\|J_A(x) - J_A(y)\|^2 + \gamma \|x - y - (J_A(x) - J_A(y))\|^2 \leq \gamma \|x - y\|^2$

Fixed points: $\text{Fix}(J_A) = \text{zer}(A)$

Applications:

- *Proximal Point Algorithm*: $x_{n+1} = J_A(x_n)$
- *Douglas-Rachford Splitting*: $\text{Fix}(T_{A,B})$ solves $0 \in Ax + Bx$

Constrained Composite Inclusions I

Problem 26.35: General multiple operator composite inclusion.

Let $z \in \mathcal{H}$, $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ be maximally monotone, I a nonempty finite index set. For each $i \in I$:

- K_i is a real Hilbert space
- $r_i \in K_i$, $B_i : K_i \rightarrow 2^{K_i}$ maximally monotone
- $L_i \in \mathcal{B}(\mathcal{H}, K_i) \setminus \{0\}$

Solve the **primal inclusion**:

$$\text{find } x \in \mathcal{H} \text{ such that } z \in Ax + \sum_{i \in I} L_i^*(B_i(L_i x - r_i))$$

and the **dual inclusion**:

$$\text{find } (v_i)_{i \in I} \in \bigoplus_{i \in I} K_i \text{ such that } (\forall i \in I) \quad -r_i \in L_i A^{-1} \left(z - \sum_{j \in I} L_j^* v_j \right) + B_i^{-1} v_i$$

Solution: Use Spingarn's method or modified Douglas-Rachford on product spaces.

Constrained Composite Inclusions II

Corollary 26.36: Under suitable qualification conditions:

$$z \in \text{ran} \left(A + \sum_{i \in I} L_i^* \circ B_i \circ (L_i \cdot -r_i) \right)$$

An iterative algorithm (Corollary 26.36):

Constrained Composite Inclusions III

$$y_{1,n} = x_n - \gamma \sum_{i \in I} L_i^* v_{i,n}$$

$$p_{1,n} = J_A(y_{1,n} + \gamma z)$$

(for each $i \in I$):

$$y_{2,i,n} = v_{i,n} + \gamma L_i x_n$$

$$p_{2,i,n} = y_{2,i,n} - \gamma(r_i + J_{-\gamma B_i}(\gamma^{-1} y_{2,i,n} - r_i))$$

$$q_{2,i,n} = p_{2,i,n} + \gamma L_i p_{1,n}$$

$$v_{i,n+1} = v_{i,n} - y_{2,i,n} + q_{2,i,n}$$

$$q_{1,n} = p_{1,n} - \gamma \sum_{j \in I} L_j^* p_{2,j,n}$$

$$x_{n+1} = x_n - y_{1,n} + q_{1,n}$$

Key Theoretical Results Summary

Main Theorems in Chapter 26:

Theorem	Result
26.1	Duality characterizations of zeros of $A+B$
26.5	Zeros of finite sums: $x = x_1 = \cdots = x_m$
26.9	Spingarn's partial inversion method converges
26.10	Douglas-Rachford splitting convergence
26.11	Proximal point algorithm weak convergence
26.16	Linear convergence under strong monotonicity
26.20	Forward-backward algorithm convergence
26.25	Forward-backward for variational inequalities

Connections to Other Chapters

Key references used:

- **Chapter 4:** Firmly nonexpansive operators and averaged operators
- **Chapter 15:** Duality, Fenchel conjugate, and dual problems
- **Chapter 16:** Properties of operators (cocoercivity, strong monotonicity)
- **Chapter 20:** Constrained optimization methods
- **Chapter 22-23:** Subdifferentials and proximal operators
- **Chapter 24:** Proximal point algorithms and special methods

Forward references:

- Applications to variational inequalities (Section 26.7)
- Stochastic algorithms (extensions of deterministic framework)

Chapter 26: Summary

Key Concepts:

- 1 **Resolvents:** $J_A = (\text{Id} + A)^{-1}$ map zeros of A to fixed points of J_A
- 2 **Splitting methods:** Douglas-Rachford and forward-backward decompose difficult problems
- 3 **Fixed point iteration:** Proximal point algorithm converges weakly to zeros
- 4 **Duality:** Solutions to $0 \in Ax + Bx$ and dual problem are characterized together
- 5 **Convergence rates:** Linear under strong monotonicity, can be superlinear with acceleration

Main Applications:

- Variational inequalities on convex sets
- Composite optimization ($\min f + g$)
- Constrained monotone inclusion problems
- Operator splitting for parallel computation

Practical Implementation:

- Choose algorithm based on operator structure
- Tune stepsizes γ for convergence rate
- Use fixed point iteration for simplicity, splitting for modularity